

From Concept to Reality: Deploying 5G Enabled Robots in Complex Forest Scenarios



Panagiotis T. Karfakis, André Araújo, Carlos A. P. Pizzino, Micael S. Couceiro, and David Portugal

Abstract With the advent of the New Radio era, we discuss the transformative impact of 5G technologies on rural applications, particularly in the fields of forestry and agriculture, where manual labor can be automated through robotics. We highlight challenges faced by robotic systems in outdoor environments, such as GNSS limitations and sensory drift, and consider the potential of integrating 5G cellular technology as a complementary localization source, improving the precision and reliability of robotic navigation. In particular, this chapter presents developments in NR5G-COOP, the multi-robot version of our framework, within a forest, exploring UAV cooperation for enhanced radio localization and assessing the outcomes of these experiments. This is currently a work in progress and we aim to explore the capabilities of our cooperative radio localization framework in forest areas, where anchor visibility is obstructed, LiDAR-inertial based systems drift and GNSS may be impaired.

Keywords 5G NR · Cooperative radio localization · GNSS-denied · Sensory drift · Multi-robot systems · Forest environments · LiDAR SLAM · UAV

P. T. Karfakis (✉) · D. Portugal

Institute of Systems and Robotics (ISR), Department of Electrical Engineering and Computers (DEEC), University of Coimbra (UC), Coimbra, Portugal

e-mail: panagiotis.karfakis@student.uc.pt

D. Portugal

e-mail: davidbsp@isr.uc.pt

A. Araújo · C. A. P. Pizzino · M. S. Couceiro

Ingeniarius Ltd., Alfena, Portugal

e-mail: andre@ingeniarius.pt

C. A. P. Pizzino

e-mail: carlos@ingeniarius.pt

M. S. Couceiro

e-mail: micael@ingeniarius.pt

1 Introduction

Rural environments host numerous applications that support primary sectors of the economy, such as forestry and agriculture. These domains often involve labor-intensive, manual tasks with significant potential for automation. As a key enabler of this transition, 5G technologies offer promising capabilities for robotic systems operating in outdoor settings (Tang et al. 2021; Nesse et al. 2023). These environments present challenges for robotic perception as they impact the sensing modalities that robotic systems rely on for navigation.

Modern sensing devices, such as LiDAR and cameras, have driven breakthroughs in robotic localization in these environments. Their fusion with inertial and GNSS sensors has further increased overall robustness. Challenges for optical-based sensing include the uniformity of forest environments, which significantly impacts vision-based methods due to lighting variations and the lack of distinctive visual cues caused by dense foliage. LiDAR sensing ensures stable performance allowing for range measurements to nearby objects and generation of detailed point clouds, which are essential for mapping and navigation in complex or unstructured environments. In dense forests and agricultural settings, unfortunately issues persist due to the canyon effect on GNSS systems, accumulated drift in inertial sensing, and the homogeneity or lack of features in the environment, which causes drift in optical-based sensing.

As a step forward, the integration of 5G network technologies, which range from 4.1–300 GHz and is inclusive of previous wireless standards, such as 4G LTE-A, 3G, WiFi and UWB, presents additional features and capabilities, such as connection speeds, ultra low latency, increased device capacity, larger bandwidth, availability and coverage (Dilli 2020). In this work, we attempt to emulate the performance of a 5G positioning service in the Sub-6 GHz frequency band in a rural setting, by deploying static UWB nodes in irregular geometry and utilizing cooperative signals onboard a team of mobile robots. We explore its feasibility as an additional standalone or complementary absolute localization source, similar to GNSS, but based on terrestrial 5G cellular stations. With this technology robotic systems would be able to utilize signals and resources from nearby base stations and infer their whereabouts, when other onboard methods are impacted or unavailable.

In this chapter, we present the design and planned developments for our NR5G-COOP algorithm in a forest environment, towards deploying radio localization capabilities within a team of UAVs and ultimately assess the performance of the cooperative radio localization method through field experiments, compared to onboard sensing modalities, based on GNSS and LiDAR technologies.

2 Robot Perception in Complex Forest Environments

In order to enable outdoor navigation of autonomous systems their perception ability needs to be ensured. For this reason, robots have utilized a number of sensing

modalities to survive the wildest conditions (Chen et al. 2025). GNSS has been the standard positioning modality at the global scale and is capable of providing centimeter level precision when combined with a terrestrial Real Time Kinematics (RTK) base station, which allows for position corrections and added precision (Tomašík et al. 2019). Another promising and yet mature alternative in robotic perception includes LiDAR and visual-based sensing paired with inertial devices, which have been widely put to test in outdoor environments, allowing for feature-based localization in the wild (Cristóvão et al. 2023).

Despite these attempts, robotic systems struggle in preserving long-term localization in forestry environments for a number of reasons, despite recent advancements in the field (Oliveira et al. 2021). These include the homogeneity of features such as branches, heavy foliage and color similarity of forestry environments. Other limitations include the weather conditions that may include fog, mist, humidity and rain that critically impact visual sensing. Even GNSS systems are affected in forestry environments as signal reception can not be ensured under tree canopies and rural canyons obstruct signals to the robot's receiver. To mitigate these issues, sensor fusion has been paramount to expand the working capability of robots in outdoor settings such as forestry and agriculture.

Sensor fusion includes the combination of sensing information to produce robust state estimates of robotic systems, strongly supporting navigation in unstructured environments (Zhong et al. 2021; Sousa et al. 2023). The advancement of sensor technology and extensive developments in methods, such as visual, LiDAR, inertial and their integration with GNSS, have demonstrated their strong capabilities throughout the past decades in a number of applications, but still long-term localization can not be guaranteed. In modern times, robotic systems utilize Simultaneous Localization and Mapping (SLAM) with outstanding results, complemented by the advancements in computational power in smaller form factors and the adoption of LiDAR and camera technologies to build metric or occupancy maps of the environment (Zhao et al. 2021; Wang et al. 2024). A complete example of a robotic swarm performing successfully SLAM in the wild forested environment was developed in (Zhou et al. 2022).

When tested in forestry environments, optical-based SLAM systems, are still prone to failure due to the accumulation of errors in state estimation, known as drift (Bresson et al. 2015). This occurs due to many factors such as lack of features, homogeneity of the environment, and overconfidence in their predicted state. The effect of drift is the gradual divergence of its localization, and consequently the generated map of the environment, from their real world values. Furthermore, the uncertainty related to the robot's position does not encompass the true pose, indicating that the estimate is unreliable and should not be depended upon in subsequent iterations, as errors will be additive.

In the case of wireless and radio localization this is achieved via range determination via the collective acquisition of signals from nearby stations or from robot team members, that have been deployed in the scenario (Mahyuddin et al. 2017). Once all meaningful information has been collected, the robot is capable of determining its pose and sharing it in similar fashion with other team members to reset errors

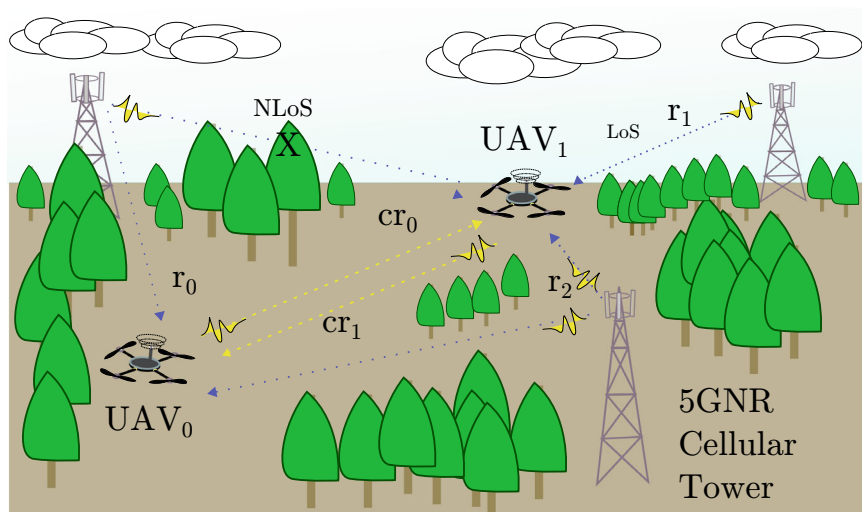


Fig. 1 Cooperative localization from 5G base station signals and other robots

in their state estimation pipelines, this is known as cooperative localization (Wang et al. 2025). An illustration of this is depicted in Fig. 1, in which 5G base stations also known as gNodeBs transmit ranging signals and UAVs share their ranging and pose information.

In cooperative localization, multiple wireless nodes (also referred to as static stations, anchors or reference nodes) are collectively utilized to determine the position of a mobile robot (Zhou et al. 2024). Unlike traditional radio localization, where a single device can only rely on the signals of a few known points, as in GNSS, cooperative localization uses the collective information and measurements between different neighboring nodes or as a network. Two types of nodes are necessary in a cooperative scenario: static stations, with known and fixed positions, which transmit and receive signals from the mobile robot; and mobile nodes, whose position needs to be tracked over time, being capable of communicating signals from/to fixed, mobile or both node types.

For cooperation to occur, static and mobile robots must share information and their measurements with each other to improve overall localization accuracy and coverage. The shared information may include range measurements, which are the estimated distance between nodes and is computed using radio localization methods such as Time of Arrival (ToA), Time Difference of Arrival (TDoA), Received Signal Strength Indicator (RSSI) or Angle of Arrival (AoA) (Guo-wei et al. 2011), or connectivity state information, availability of nodes and their types, LoS/NLoS conditions and channel state information, such as RSSI. Ultimately the information from all available nodes can be grouped and used to refine position estimates of each robotic system.

Measurements in the form of signals initially transmitted by static stations are received by mobile robots and may include signal strength, time delays and/or angles.

These measurements and their associated uncertainties are then exchanged between the cooperating nodes of the multi-robot system, along with their own set of radio information. This can be done centrally, where all data is sent to a central processor over the network or in distributed fashion, where each node exchanges information with their neighbors nodes, without the need of a central processor. Algorithms such as trilateration, multilateration (Zhou et al. 2009), Kalman filters (Ting Goh et al. 2018), particle filters (Iatsyshyn 2025), factor graphs (WU et al. 2022), can be applied to these shared measurements to estimate the position of the mobile robot. Cooperative localization enables the production of a position estimate even when a mobile robot cannot communicate directly with enough static stations. In such cases, it can rely on information relayed or refined by other mobile nodes, including areas with irregular geometry or outside the coverage envelope of static nodes.

By combining multiple measurements and leveraging the redundancy of information, cooperative localization architectures can significantly improve localization accuracy, especially in challenging environments such as urban canyons and GPS-denied areas, where direct LoS or strong signals may be limited (Han et al. 2024). A mobile robot may not always be in direct range of enough stationary nodes to localize itself independently, which might lead to ambiguity of the estimated solution. Collaboration enables localization even in areas with sparse anchor deployments, as the information can be relayed over the network or among members. Thus, the system becomes more resilient to individual sensor failures or noisy measurements. If one measurement is poor, other cooperative measurements can compensate and improve the state estimation pipeline of each robot of the team. In some cases, the need for a dense deployment of high-precision (and expensive) anchor nodes can be reduced as less accurate nodes can contribute to the overall solution depending on localization requirements. Cooperative multi-robot systems, are able to adapt to changes in the environment or network topology as they continuously update their estimates based on new information shared among the members of the team.

A number of challenges in cooperative localization frameworks persist and need to be mitigated (Meng et al. 2025). Combining and processing a large number of measurements from multiple nodes, especially in real time for a single mobile robot, can be computationally intensive. This may be a problem for devices with limited resources. Many range-based localization techniques (such as ToA/TDoA) require precise time synchronization between transmitting and receiving nodes in order to determine the pseudo ranges from static nodes. Maintaining this synchronization in a distributed architecture can be a challenge, unless a common clock reference is available to all members. The exchange of measurements and also position estimates between cooperating nodes generates communication traffic that can consume valuable bandwidth and energy, especially in large 5G, in Internet of Things (IoT) networks or in robotic swarms (Rajendra et al. 2022). In real-world environments, signals are obstructed by obstacles, resulting in multipath fading and ambiguous, NLoS propagations, which distort measurements and introduce significant noise and interference, that cooperative localization algorithms must handle. As the number of static stations and mobile targets increases, maintaining coordination and efficiency

of communications and computations becomes a major challenge that undermines localization performance.

To mitigate the limitations of onboard sensing methods such as drift and GNSS unavailability, we believe that a terrestrial radio localization source based on 5G cellular signals, can serve as a complementary localization source when fused with other sensing information in complex outdoor settings. In our previous work, we introduced NR5G-SAM (Karfakis et al. 2023), a SLAM framework based on 5G New Radio signals from multiple terrestrial base stations. This proof of concept implementation utilizes 5G radio stations and the positioning service to localize a mobile robot in an outdoor agricultural setting. The algorithm makes use of Position Reference Signals (PRS) for ranging, as well as channel state information, such as RSSI for SLAM using factor graphs, while performing radio environmental mapping to provide robustness and long term operation.

In this chapter, we discuss the expansion of NR5G-SAM to a cooperative multi-robot localization system, which we refer to as NR5G-COOP. The cooperative method utilizes ranging signals, signal strength and multilateration to determine the positions of UAVs. At the moment, we are working towards deploying the method in a complex real-world forest landscape to showcase its abilities, considering the numerous challenges involved and further assessing its performance against GNSS positioning and LiDAR-based odometry and mapping.

3 Methodology

We have designed NR5G-COOP framework as a cooperative radio localization approach for multi-robot systems (see Fig. 2), which utilizes radio information such as the pseudo range between static and mobile nodes, RSSI and other information shared among robots. This information includes the range information and the pose estimated independently from each UAV. Each UAV has its own GNSS time receiver and its own clock reference, which is common across all platforms. Moreover, the station locations are known and referenced in the GNSS coordinate frame.

Transmitted radio signals include error from multiple sources that are due to propagation conditions, interferences, obstructions and reflections, and scattering conditions. The overall radio signals transmitted in our cooperative framework can be seen in Fig. 3. The general equation of a radio signal is:

$$s_i = r_i + n_i, \quad (1)$$

where s_i is the signal received from a station or robot and n_i is the noise present in the measurement which for simplicity is assumed to follow a Gaussian distribution. The state information obtained by the wireless ranging system is the ranging information:

$$r_i = \frac{t_{rx} - t_{tx}}{c}, \quad (2)$$

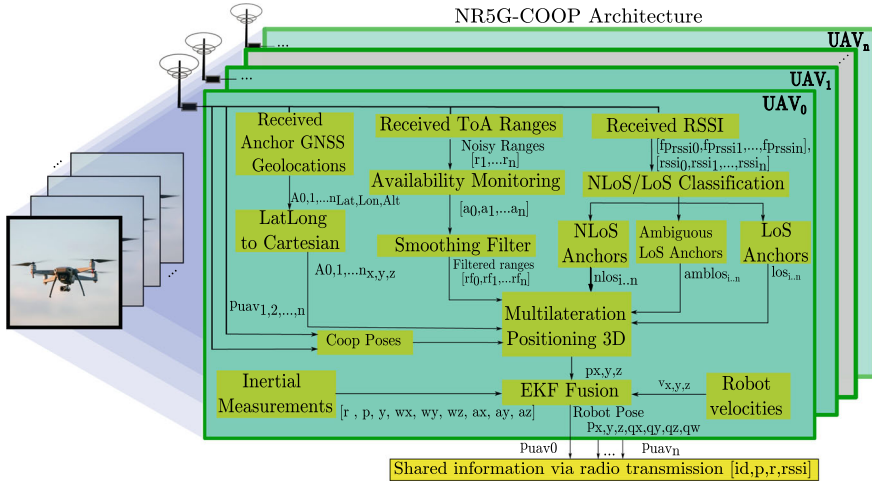


Fig. 2 The NR5G-COOP architecture, which utilizes radio signals from static nodes and mobile nodes along with shared information from multiple UAVs, such as their own pose and range information. Each UAV has its own state estimation pipeline using an EKF that fuses IMU and velocity information

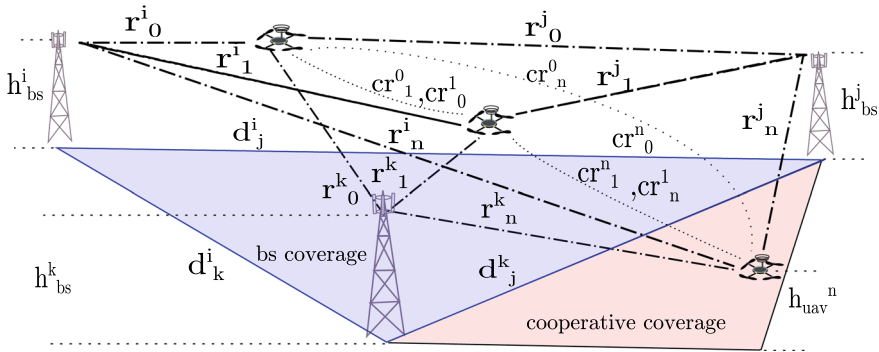


Fig. 3 Distances d_n^i , ranges r_n^i and cooperative ranges cr_n^i between stations and mobile robots. The height of the stations h_{bs}^i and altitude of the UAVs h_{uav}^i are used for the multilateration estimation process

where r is the range from the i th anchor, t is the time of reception rx and transmission tx , and c is the speed of light constant (set as 3×10^8). These ranges are used in a multilateration inference module as described in Karfakis et al. (2023) to generate the position estimates from anchor nodes with known positions. The anchor locations are assumed to be known as in 5G cellular systems have known GNSS installation locations which are usually publicly available from telecommunication operators.

In forests where trees are predominately dense signal dropouts are highly likely, as well as obstructions from the dense foliage. These situations are handled in this work

using the RSSI thresholding for each connection to classify whether the information should be used or not. If the loss or drop is larger than 6 dBm on average the information obtained is not considered reliable and most likely its a non line of sight condition. Therefore in these scenarios we will attempt to rely on cooperative information from the robot team members.

Next in the state estimation pipeline is the determination of the range from fellow robot teammates or agents. This condition differs from the anchor case in that the UAV locations are relevant between members and are generated independently by their own perception layer. The condition that needs to be met is that the range information and the robot's location is transmitted in timely fashion between members or through the network. In our case, we have chosen not to utilize the network-based approach due to communication delays and obstructions of signal reception in a forest. Instead, the relative inter-robot ranges are computed using the robot's relative position as follows:

$$cr_{ij} = |\hat{p}_i - \hat{p}_j| + e_{ij}, \quad (3)$$

where cr_{ij} is the cooperative range from UAV with identifier i to UAV j , and their estimated positions $\hat{p}_i$, $\hat{p}_j$ and the inherent error e_{ij} .

After the ranges have been estimated we proceed with the optimization. In our case, we utilize a Genetic algorithm which has been validated previously in Karfakis et al. (2022). The optimization and prediction of the state estimate combines the locations of the static anchors and the mobile robots using the following set of equations:

$$\bar{x} = bs_{x_i} - \hat{x}_i, \quad (4)$$

$$\bar{y} = bs_{y_i} - \hat{y}_i, \quad (5)$$

$$\bar{z} = bs_{z_i} - \hat{z}_i, \quad (6)$$

$$\hat{p}_i = \sqrt{\bar{x}^2 + \bar{y}^2 + \bar{z}^2} - r_i, \quad (7)$$

$$\hat{p}_{ij} = \sqrt{\bar{x}^2 + \bar{y}^2 + \bar{z}^2} - cr_{ij}, \quad (8)$$

in which bs_{x_i} , bs_{y_i} , bs_{z_i} are the static node locations and $\hat{x}$, $\hat{y}$, $\hat{z}_i$ the estimated state estimates from the iterative least squares multilateration. The pose $\hat{p}_{ij}$ is the result of the cooperative positioning between mobile and static nodes. The expanded form of the formula for computing the position of the robot is the following,

$$p_{x,y,z} = \arg \min || \sum_{i=1}^N \left(r_i - \sqrt{(x_i - x_{bs})^2 + (y_i - y_{bs})^2 + (z_i - z_{bs})^2} \right)^2 + \sum_{i=1}^N \left(cr_{ij} - \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2} \right)^2 ||, \quad (9)$$

where the robot position $p_{x,y,z}$ is estimated through the static station ranges r_i and mobile cooperative ranges cr_{ij} along with the known static and the estimated mobile node positions p_i and $\hat{p}_{ij}$ at the given timestep.

In addition to the state estimation layer, we have introduced a dynamic selection scheme based on RSSI thresholding. We obtain RSSI measurements along each timestep, complementary to the ranges and we perform an assessment of signal quality. The stations with stronger values are sorted and then multilateration is performed to estimate the robot's state, according to their associated identifiers. This assumption is based on the relationship between signal power and distance between the transmitter and receiver, in which higher RSSI measurements imply closer proximity to available nodes.

Once we obtain the raw positioning results from the multilateration process, we utilize an Extended Kalman Filter (EKF) to track each robot separately and also fuse inertial measurements, from an Inertial Measurement Unit (IMU) and robot velocities, to produce a 6-DoF pose. Sensor fusion is necessary as radio localization approaches based on multilateration, do not contain heading information and estimation of velocities from radio signals contain high uncertainty in Sub6-GHz frequencies. Our proposed approach is a terrestrial radio localization framework. We follow the modeling of the filter and process configuration according to Moore and Stouch (Moore and Stouch 2014). The overall state estimation process for each robot is modeled as a nonlinear dynamic system of the form:

$$x_k = f(x_{k-1}) + w_{k-1}, \quad (10)$$

where x_k is the robot's system state vector, in our case the position, orientations, velocities and accelerations,

$$x_k = [p_{x,y,z}, q_{x,w,z,w}, v_{x,y,z}, \omega_{x,y,z}, \alpha_{x,y,z}], \quad (11)$$

at time k , f is a nonlinear state transition function, and w_{k-1} is the process noise which is assumed to follow a Gaussian distribution and inclusive within measurements. Rotational values are expressed in quaternions. The robot state measurements are expressed according to the following form:

$$z_k = h(x_k) + u_k, \quad (12)$$

where z_k is the measurement at time k , h is a nonlinear sensor model that maps the state into measurement space, and u_k is the measurement noise, which is assumed to

be Gaussian. During the prediction step, the filter projects the current state estimate and error covariance forward in time:

$$\hat{x}_k = f(x_{k-1}), \quad (13)$$

$$\hat{P}_k = F P_{k-1} F^T + Q. \quad (14)$$

For our application, f is a the kinematic model of the robot. The error covariance of the estimation, P , is projected via F , the partial derivative or Jacobian of f , and then is affected by the dynamic process noise covariance Q . During the correction step of the filter, a new measurement z_k is compared with the predicted state estimate in order to refine it. This step aims to reduce the uncertainty in the state estimate by comparing the predicted state with the observed measurement using the following set of equations:

$$K = \hat{P}_k H^T (H \hat{P}_k H^T + R)^{-1}, \quad (15)$$

$$x_k = \hat{x}_k + K (z_k - H \hat{x}_k), \quad (16)$$

$$P_k = (I - KH) \hat{P}_k (I - KH)^T + K R K^T, \quad (17)$$

where K is the Kalman gain, H is the observation matrix, R is the measurement covariance, and $\hat{P}_k$ is the predicted error covariance.

Finally the EKF operates by predicting the robot pose using its dynamic motion model and then refines this prediction with actual sensor data, from the multilateration and IMU accounting for uncertainties in both the model and the measurements. This iterative cycle of prediction and correction steps, involves linearizing the non-linear motion and measurement models around the current estimate, hence it enables the filter to maintain an accurate and smooth estimate of the pose. Our aim is to evaluate the robustness and localization performance, despite sensor noise and temporary data dropouts of radio connectivity in forest environments due to signal obstructions from dense foliage.

4 Evaluation Planning and Experimental Setup

To support the validation of our framework in a forested area we have conducted a field test in an eucalyptus forest on the outskirts of Porto, in Alfena, Portugal. The Linktrack (P and P-B series) UWB nodes are deployed as nodes that contain unique identifiers and roles. We have installed three anchor nodes at relevant locations to emulate the signal reception of cellular signals in a rural environment. This is a highly limiting scenario for multilateration approaches. The distances between them were arbitrarily selected and occlusions were taken into consideration, which makes the scenario more challenging and promotes the realistic dynamics of rural environments. Their height was set in similar fashion and differ significantly among



Fig. 4 The forest scenario of the multi-robot application and the planned trajectories of the UAVs

each other, which inherently will affect radio localization performance in the other two dimensions. The coverage envelope in this scenario is not guaranteed over all the application area and contains multiple tree obstructions, which are present in the test site as can be seen in Fig. 4.

The robotic systems used in this experiment are three Scout v3 UAVs, that are equipped with a 3D LiDAR, IMU, and GNSS-RTK receiver (see Fig. 5a). The UWB nodes were added to receive measurements from the static nodes and mobile nodes of the UAVs. Evaluations of the performance of various visual and LiDAR-based algorithms in forestry environments, have shown that LiDAR solutions generally exhibited superior results compared to visual alternatives and under illumination changes throughout the day. To support the reporting of the localization performance an RTK base station has been installed and all localization information are referenced in its coordinate reference system (see Fig. 5b), to assist ground truth reporting and comparison. Additionally with LiDAR mapping the environment, the representation can be geographically referenced and stored as 3D point clouds.

Collaborative SLAM capabilities have been achieved by integrating two frameworks, namely DCL-SLAM (Distributed Collaborative Localization and Mapping) (Zhong et al. 2023) and the front-end module based on LIO-SAM (LiDAR Inertial Odometry via Smoothing and Mapping) (Shan et al. 2020). The integration of these two algorithms is called DCL-LIO-SAM.



Fig. 5 **a** The sensor kit onboard the Scout UAV. **b** The deployment of the UAVs along with the RTK base station in the background

DCL-SLAM is a framework developed for real-time, distributed localization and mapping in collaborative multi-robot systems. It enables multiple drones to operate cooperatively to construct a detailed map of the environment while simultaneously localizing themselves within it.

To integrate DCL-SLAM into our UAV platform, we take advantage of the modular and extensible architecture of the MRS UAV system (Baca et al. 2021), a full-stack open-source multi-rotor UAV platform designed for realistic simulation, real-world deployment, and reproducible research.

This framework provides a robust interface for incorporating external localization sources through its multi-frame state estimation module, which supports concurrent hypotheses via a bank-of-filters approach. By treating DCL-SLAM as an additional localization input, we fuse its position and heading estimates alongside other onboard sensor data and estimators. The arbiter mechanism within the estimator selects the most reliable hypothesis in real-time, ensuring seamless transitions between localization sources. This design enables DCL-SLAM to enhance situational awareness and cooperative localization capabilities, particularly in multi-agent scenarios and GPS-denied environments, without disrupting the core control or navigation pipelines.

5 Evaluation Metrics

For the evaluation of the state estimation performance overall we use the root mean squared error (RMSE), which is expressed as:

$$RMSE = \sqrt{(\hat{x}_i - x_{gt_i})^2 + (\hat{y}_i - y_{gt_i})^2 + (\hat{z}_i - z_{gt_i})^2}, \quad (18)$$

where the inferred 3D coordinates $[\hat{x}_i, \hat{y}_i, \hat{z}_i]$ are generated by each localization method. These are subtracted from the GNSS-RTK coordinates at each timestep $[x_{gt_i}, y_{gt_i}]$, which serves as ground truth reference. The z_{gt_i} parameter is given by the onboard altimeter. Additionally to this metric, we compute the standard deviations of each method using:

$$\sigma = \sqrt{\sum_{i=1}^n \frac{(\hat{x}_i - x_{gt_i})^2 + (\hat{y}_i - y_{gt_i})^2 + (\hat{z}_i - z_{gt_i})^2}{n - 1}}, \quad (19)$$

where n is the length of the estimated sets compared to the ground truth.

6 Results and Discussion

In this experiment we recorded the estimations from GNSS-RTK and DCL-SLAM to allow us to assess in the future the performance of NR5G-COOP localization in the forest environment where the weather conditions were fairly windy making things more challenging.

Figure 6 presents the trajectories of all the robots based on the GNSS-RTK positioning system, that we have deployed in the field. This is used as ground truth for comparison with the estimated localization from the NR5G-COOP and DCL SLAM frameworks. Despite seeming continuous, the GNSS measurements are not always in fixed status, which is common in forest environments, as the trees obstruct the base station visibility with the UAVs temporarily during flight. In such environments, estimations from single GNSS receivers will jump or even may be unavailable temporarily. The complete scenario can be assessed from the pointcloud map of Fig. 7. Fortunately, we have made provisions in the waypoint selection to maintain the fix status throughout the scenario in order to have a ground truth reference and fly the UAVs with confidence, avoiding crashes (Fig. 8).

The performance of DCL-SLAM suffered significant drift during flights as errors accumulated over time, causing the UAVs to lose their whereabouts. This phenomenon occurs in higher altitudes between trees, as well as in areas with no trees and near the canopies. The DCL-SLAM trajectories are depicted in Fig. 9, and despite having overlapping trajectories and an active loop closure module, DCL-SLAM failed to localize UAV 2 for the duration of its flight, causing its estimation of local-

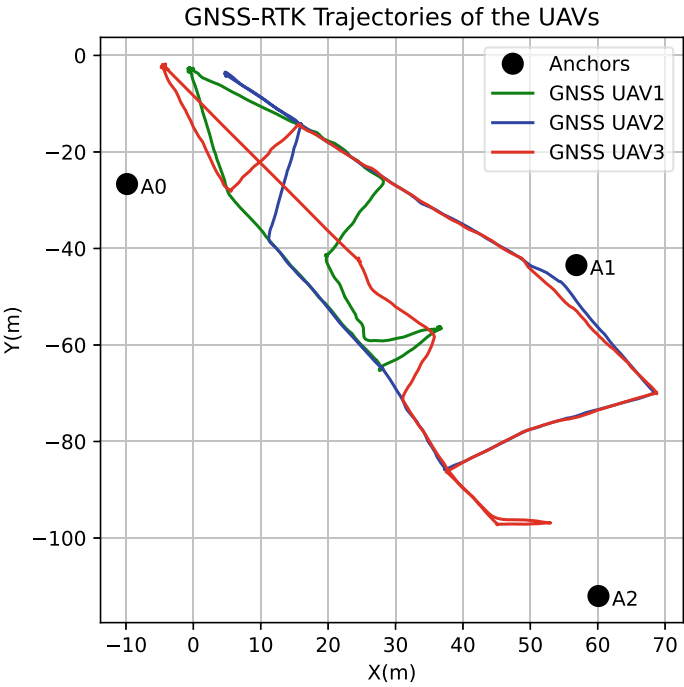


Fig. 6 The ground truth localization of the UAVs given by GNSS-RTK

ization to drift. Another interesting fact is that the accumulation of drift during flight has affected the orientation of the LiDAR SLAM approach as well, compared to ground truth trajectories of Fig. 6.

DCL-SLAM has diverged in certain scenarios. Specifically, the divergence occurs when the drones ascend to high altitudes. At these altitudes, the LiDAR SLAM method, at times is unable to perceive and model the environment accurately, due to the scale and lack of sufficient features, nor maintain reliable localization (see Fig. 8). As a result, the SLAM algorithm struggles to compute accurate pose estimates, leading to drift and eventual divergence in the mapping and localization output. This is a common challenge in field robotics, especially optical and inertial SLAM systems operating in outdoor environments. Nevertheless, it is important to highlight that this behavior occurs under conditions that are outside the intended operational experiment. We recognize that comparing or evaluating SLAM performance under these exceptional conditions may not provide a fair assessment of the system’s actual capabilities under nominal scenarios in a forest.

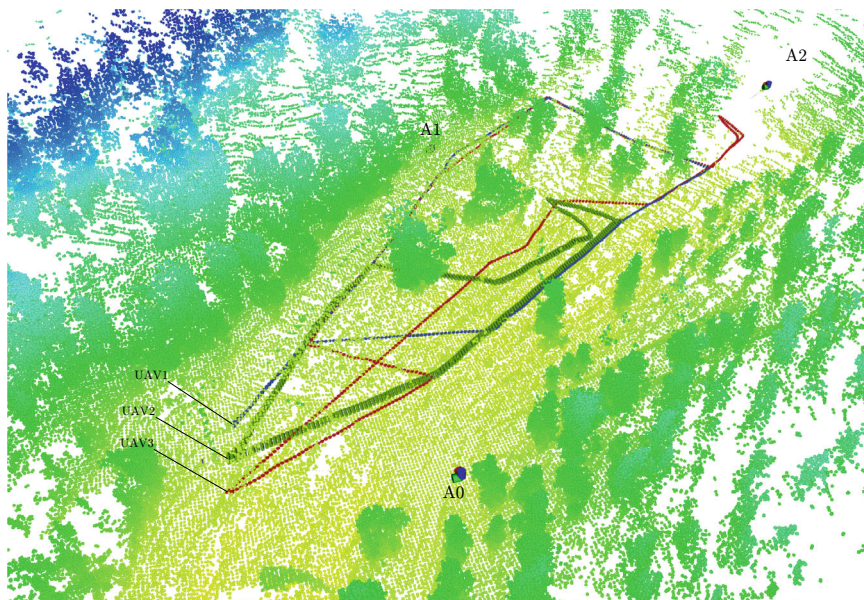


Fig. 7 The mapped forest area with the UAVs along with their GNSS trajectories. The anchor locations are labeled in black and the color encoding of the pointcloud is based on the z-axis

Regarding NR5G-COOP, we are currently performing the testing phase of the algorithm to obtain results. Unfortunately, we would need to perform the test again as during the data collection operation we suffered a connectivity issue for the UWB node of UAV2, which limits the estimation process to have a fair comparison along the complete trajectory. We will soon present our findings within this section. From the limited data we have acquired, areas exist where the signals are unavailable due to the tree lines blocking or deteriorating the range measurements. Along the trajectories of the UAVs we have witnessed cases where there are not enough UWB nodes, as multilateration requires a minimum of three to generate an estimation. We have also encountered areas, where indeed UAVs with cooperative ranges provide support to other robots when static nodes are unavailable even outside the working static station envelope. We expect to perform the radio information collection and report our findings.

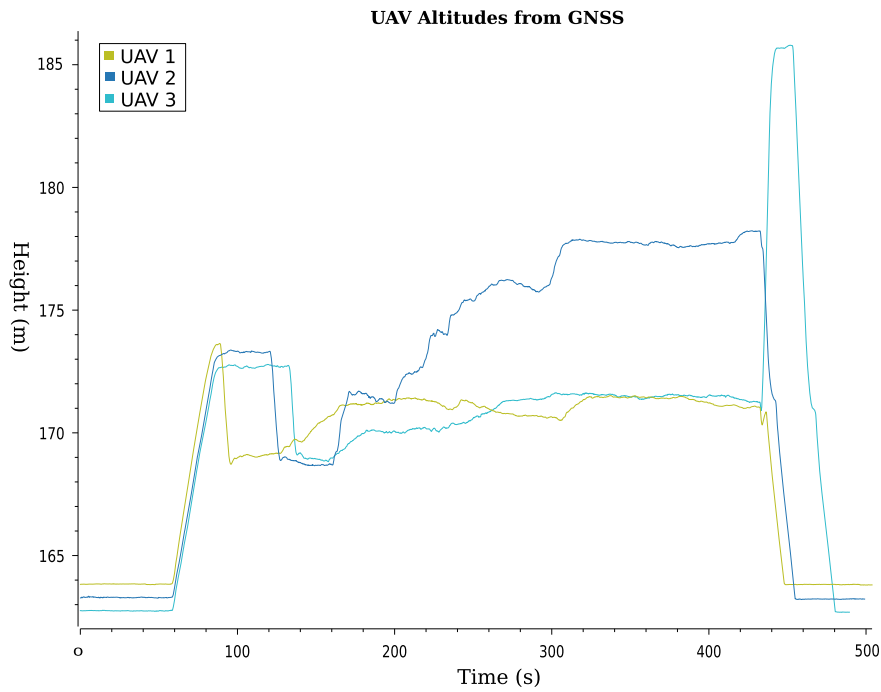


Fig. 8 The altitude of the UAVs throughout the experiment

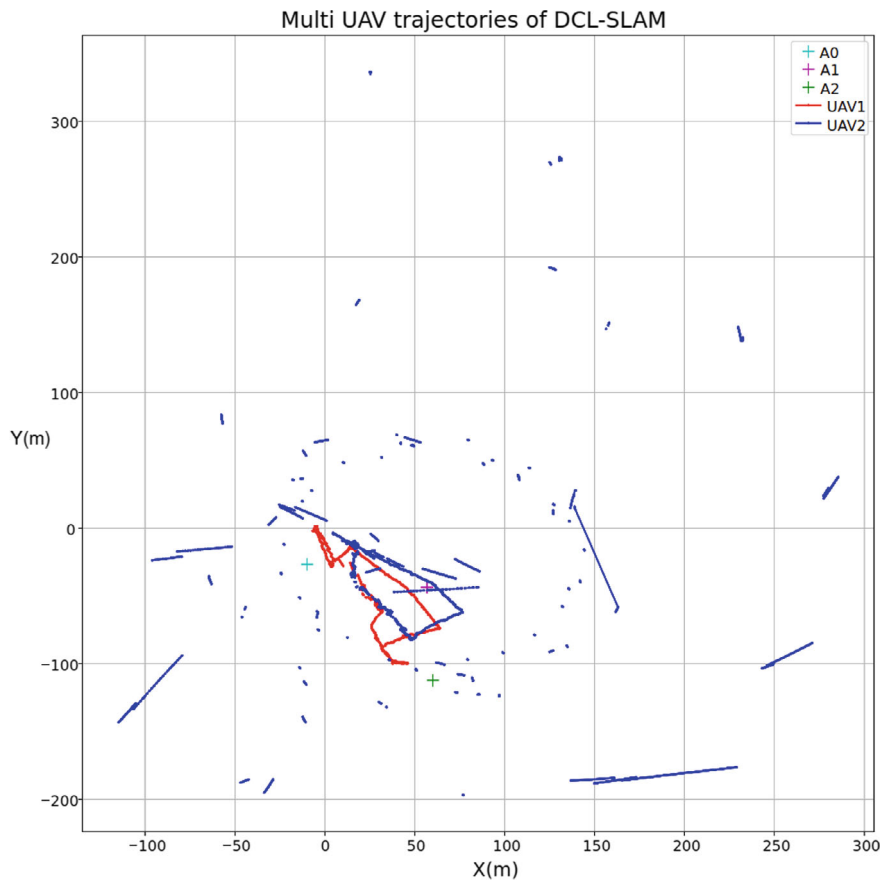


Fig. 9 The localization of the DCL-SLAM multi-robot system and the overall drift problem

7 Conclusion

In this work we have presented the challenges in robotic perception and robot localization in forest environments. We have introduced the NR5G-COOP approach based on terrestrial and onboard cooperative ranging. The ultimate goal of our work is to provide an alternative, cooperative radio localization source capable of supporting multi-robot systems in forestry and outdoor applications. Our evaluation which is a work in progress is foreseen to present concrete results despite onboard sensing failures due to drift, as optical-based sensing might be impaired or GNSS unavailable, under the presence of foliage and rural canyons.

Acknowledgements This work was funded by the European Union's Horizon 2020 research and innovation programme 5GSmartFact under the Marie Skłodowska-Curie grant agreement ID 956670. David Portugal also acknowledges the Portuguese Science Agency "Fundação para a Ciência e a Tecnologia" (FCT) for support under the Scientific Employment Stimulus 5th Edition, contract reference 2022.05726.CEECIND.

References

- Baca T, Petrlik M, Vrba M, Spurny V, Penicka R, Hert D, Saska M (2021) The MRS UAV system: pushing the frontiers of reproducible research, real-world deployment, and education with autonomous unmanned aerial vehicles. *J Intell Robot Syst* 102
- Bresson G, Aufrère R, Chapuis R (2015) Improving SLAM with drift integration. In: 2015 IEEE 18th international conference on intelligent transportation systems, pp 2700–2706
- Chen W, Chi W, Ji S, Ye H, Liu J, Jia Y, Yu J, Cheng J (2025) A survey of autonomous robots and multi-robot navigation: perception, planning and collaboration. *Biomimetic Intell Robot* 5:100203. <https://www.sciencedirect.com/science/article/pii/S2667379724000615>
- Cristóvão MP, Portugal D, Carvalho AE, Ferreira JF (2023) A LiDAR-Camera-Inertial-GNSS apparatus for 3D multimodal dataset collection in woodland scenarios. *Sensors*. 23(15). <https://doi.org/10.3390/s23156676>
- Dilli R (2020) Analysis of 5G wireless systems in FR1 and FR2 frequency bands. In: 2020 2nd international conference on innovative mechanisms for industry applications (ICIMIA), pp 767–772
- Guo-Wei Z, Jia-Yu W, Qiu-Hong L (2011) Localization of robots based on wireless sensor networks. In: 2011 international conference on consumer electronics, communications and networks (CEC-Net), pp 5027–5030
- Han Z, Ding H, Han L, Ma L, Zhang X, Lou M, Wang Y, Jin J, Wang Q, Liu G, Wang J (2024) Cellular network based multistatic integrated sensing and communication systems. *IET Commun* 18:1878–1888. <https://ietresearch.onlinelibrary.wiley.com/doi/abs/10.1049/cmu2.12732>
- Iatsyshyn V (2025) Particle filter application to radio-based range-only UAV self-localization. *Developments in information and knowledge management systems for business applications* 8:149–165. https://doi.org/10.1007/978-3-031-80935-4_8
- Karfakis PT, Couceiro M, Portugal D (2023) NR5G-SAM: a SLAM framework for field robot applications based on 5G new radio. *Sensors* 23. <https://www.mdpi.com/1424-8220/23/11/5354>
- Karfakis P, Couceiro M, Portugal D, Antunes CA (2022) Comparative study of mobile robot positioning using 5G-NR. In: IEEE international conference on robotics and automation: IFRRIA

- workshop on innovation in forestry robotics: research and industry adoption, May 23–27, 2022, Philadelphia (PA), USA. <https://openreview.net/forum?id=Sn4xddVNEfc>
- Mahyuddin M, Isa A, Zin M, Afifah Maheeran AH, Manap Z, Ismail MK (2017) Overview of positioning techniques for LTE technology. *J Telecommun Electron Comput Eng* 9:43–50
- Meng K, Masouros C, Petropulu A, Hanzo L (2025) Cooperative ISAC Networks: opportunities and challenges. *IEEE Wirel Commun* 32:212–219
- Moore T, Stouch DA (2014) Generalized extended Kalman filter implementation for the robot operating system. In: *Proceedings of the 13th international conference on intelligent autonomous systems (IAS-13)*
- Nesse P, Nielsen N, Davidson S, Gillund L, Awan M (2023) 5G in rural forest enables real time decision support and new remote operation solutions. In: *2023 3rd international conference on electrical, computer, communications and mechatronics engineering (ICECCME)*, pp 1–6
- Norrddine A (2015) An algebraic solution to the multilateration problem
- Oliveira LFP, Moreira AP, Silva MF (2021) Advances in forest robotics: a state-of-the-art survey. *Robotics* 10(2):53. <https://doi.org/10.3390/robotics10020053>
- Rajendra M, Patra M, Srinivasan M (2022) Optimal rate and distance based bandwidth slicing in UAV assisted 5G networks. In: *ICC 2022—IEEE international conference on communications*, pp 1–6
- Shan T, Englot B, Meyers D, Wang W, Ratti C, Rus D (2020) LIO-SAM: tightly-coupled lidar inertial odometry via smoothing and mapping. In: *2020 IEEE/RSJ international conference on intelligent robots and systems (IROS)*, pp 5135–5142
- Sousa R, Sobreira H, Moreira A (2023) A systematic literature review on long-term localization and mapping for mobile robots. *J Field Robot* 40:1245 1322. <https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.22170>
- Tang Y, Dananjayan S, Hou C, Guo Q, Luo S, He Y (2021) A survey on the 5G network and its impact on agriculture: challenges and opportunities. *Comput Electron Agric* 180:105895. <https://www.sciencedirect.com/science/article/pii/S0168169920331008>
- Ting Goh, S., (Reza) Zekavat S, Abdelkhalik O (2018) An introduction to Kalman filtering implementation for localization and tracking applications. In: *Handbook of position location*, pp 143–195. <https://onlinelibrary.wiley.com/doi/abs/10.1002/9781119434610.ch5>
- Tomašík J, Mokroš M, Surový P, Grznárová A, Merganič J (2019) UAV RTK/PPK method—an optimal solution for mapping inaccessible forested areas? *Remote Sens* 11. <https://www.mdpi.com/2072-4292/11/6/721>
- Wang X, Li X, Yu H, Chang H, Zhou Y, Li S (2024) GIVL-SLAM: a robust and high-precision SLAM system by tightly coupled GNSS RTK, inertial, vision, and LiDAR. *IEEE/ASME Trans Mechatron*, 1–12
- Wang C, Yu C, Xu X, Gao Y, Yang X, Tang W, Yu S, Chen Y, Gao F, Jian Z, Chen X, Gao F, Zhou B, Wang Y (2025) Multi-robot system for cooperative exploration in unknown environments: a survey. <https://arxiv.org/abs/2503.07278>
- Wu X, Xiao B, Wu C, Guo Y, Li L (2022) Factor graph based navigation and positioning for control system design: a review. *Chin J Aeronaut* 35:25–39. <https://www.sciencedirect.com/science/article/pii/S1000936121002703>
- Zhang Q, Wang J, Jin C, Ye J, Ma C, Zhang W (2008) Genetic algorithm based wireless sensor network localization. In: *2008 fourth international conference on natural computation*, vol 1, pp 608–613
- Zhao S, Zhang H, Wang P, Nogueira L, Scherer S (2021) Super odometry: IMU-centric LiDAR-visual-inertial estimator for challenging environments. In: *2021 IEEE/RSJ international conference on intelligent robots and systems (IROS)*, pp. 8729–8736
- Zhong S, Qi Y, Chen Z, Wu J, Chen H, Liu M (2023) DCL-SLAM: A distributed collaborative LIDAR SLAM framework for a robotic swarm. *IEEE Sens J*. <https://doi.org/10.1109/JSEN.2023.3345541>

- Zhong X, Li Y, Zhu S, Chen W, Li X, Gu J (2021) LVIO-SAM: a multi-sensor fusion odometry via smoothing and mapping. In: 2021 IEEE international conference on robotics and biomimetics (ROBIO), pp 440–445
- Zhou Y (2009) An efficient least-squares trilateration algorithm for mobile robot localization. In: 2009 IEEE/RSJ international conference on intelligent robots and systems, pp 3474–3479
- Zhou B, Liu Z, Su H (2024) 5G networks enabling cooperative autonomous vehicle localization: a survey. *IEEE Trans Intell Transp Syst* 25:15291–15313
- Zhou X, Wen X, Wang Z, Gao Y, Li H, Wang Q, Yang T, Lu H, Cao Y, Xu C, Gao F. Swarm of micro flying robots in the wild. *Sci Robot* 7:eabm5954. <https://www.science.org/doi/abs/10.1126/scirobotics.abm5954>